A UNIQUE FACTORIZATION DOMAIN IS INTEGRALLY CLOSED IN ITS FRACTION FIELD

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Allowed standard prerequisites (editorial).

- Writing a fraction field element as a unit times signed prime powers. Used in the counted proof at lines 29825-29827. Encodes the element by its exponents at the irreducible elements, which is what has to be shown non-negative.
- Uniqueness of factorization in a unique factorization domain. Contradicted at the end of the counted proof (lines 29833-29835). Forbids the divisibility by an irreducible that the integral equation produces.

Lemma 0.1. *A unique factorization domain is normal.*

Proof. Let R be a UFD. Let x be an element of the fraction field of R which is integral over R. Say $x^d - a_1x^{d-1} - \ldots - a_d = 0$ with $a_i \in R$. We can write $x = up_1^{e_1} \ldots p_r^{e_r}$ with u a unit, $e_i \in \mathbf{Z}$, and $p_1, \ldots, p_r$ irreducible elements which are not associates. To prove the lemma we have to show $e_i \geq 0$. If not, say $e_1 < 0$, then for $N \gg 0$ we get

$$u^d p_2^{de_2+N} \ldots p_r^{de_r+N} = p_1^{-de_1} p_2^N \ldots p_r^N (\textstyle\sum_{i=1,\ldots,d} a_i x^{d-i}) \in (p_1)$$

which contradicts uniqueness of factorization in R. □

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